

Below is the **most complete list of login/authentication feature implementations** used in software development (from basic to modern enterprise). I'm including **real-time use cases + examples**, in a **single master table**.

✓ **Master Table — All Login / Authentication Implementations (With Real-Time Examples)**

#	Login Type / Implementation	How it works (short)	Where used in real-world	Best for	Example apps
1	Username + Password (Basic Login)	User enters username & password	Almost every application	Simple apps, internal portals	College ERP, Admin dashboards
2	Email + Password Login	Email is unique ID + password	Most modern apps	Standard consumer apps	Facebook, Amazon
3	Mobile Number + Password Login	Mobile used as ID + password	India-based services	Mobile-first apps	Local apps, small fintech
4	PIN-based Login	Numeric PIN instead of password	Banking apps, mobile apps	Short secure auth	Paytm PIN, PhonePe PIN
5	OTP Login (SMS OTP)	OTP sent via SMS	Banking, fintech, ecommerce	Passwordless login	Flipkart OTP login
6	OTP Login (Email OTP)	OTP sent to email	SaaS products, web apps	Web verification	Gmail verification
7	OTP + Password (2-step login)	Password + OTP	Finance, sensitive apps	Higher security	Net banking
8	Magic Link Login	Email link logs user in	SaaS apps	Passwordless SaaS	Slack magic link
9	Passwordless Login (OTP only)	No password stored	Modern apps	Simple onboarding	WhatsApp OTP
10	One-Time Login Code	Code shown in browser, type	Smart TVs, consoles	Device login	YouTube TV login

		in app			
11	Social Login (Google OAuth)	Login using Google OAuth	Apps/web	Faster user onboarding	Many apps "Sign in with Google"
12	Social Login (Facebook OAuth)	Login via Facebook	Consumer apps	Social onboarding	Spotify
13	Social Login (Apple Sign-in)	Apple authentication	iOS apps	Apple ecosystem	iOS apps
14	Enterprise SSO (SAML)	Company identity provider verifies	Corporate apps	Enterprise orgs	Microsoft AD SSO
15	Enterprise SSO (OAuth2 / OIDC)	SSO using modern standards	SaaS platforms	Enterprise SaaS	Okta SSO
16	LDAP Login	Auth against LDAP server	Companies/Universities	Internal systems	HRM/ERP portals
17	Active Directory Login	Microsoft AD-based auth	Enterprises	Office environments	Intranet portals
18	Biometric Login (Fingerprint)	Fingerprint unlock	Mobile apps	High usability & security	Banking apps
19	Biometric Login (FaceID/Face Unlock)	Face unlock login	Mobile/modern devices	High security login	iPhone apps
20	Device-based Login	Trusted device unlock	Secure apps	Device trust model	Google device trust
21	Remember Me Session Login	Cookie-based long session	Web apps	Convenience	Most websites
22	Session-based Auth	Server keeps session id	Web apps	Traditional backend	Django/Spring login

	(Cookie)				
23	Token-based Auth (JWT)	Client stores JWT token	APIs/mobile apps	Microservices/API	Most REST APIs
24	Refresh Token Auth	Short access token + refresh token	Secure API architecture	Production authentication	Google APIs style
25	Access Token + Scope login	Token contains permissions	Enterprise systems	RBAC/permission apps	Admin panels
26	API Key Authentication	Secret key in header	Developer APIs	Simple machine auth	Maps API, weather APIs
27	HMAC Signature Auth	Request signed with secret	Banking APIs	Strong request integrity	AWS request signing
28	mTLS Authentication	Mutual certificates (client+server)	Secure enterprise networks	Zero-trust secure backend	Banking systems
29	Login via QR Code	Scan QR from mobile	Web apps + mobile integration	Fast and secure login	WhatsApp Web
30	Login with Security Questions	Answer stored questions	Old systems	Legacy	Bank portals (old)
31	Captcha-based Login	Captcha prevents bot login	Web	Prevent brute-force	Many sites
32	Multi-Factor Authentication (MFA)	Password + OTP/biometric/app	Finance, enterprise	High security	Banking, GitHub
33	Authenticator App OTP (TOTP)	Authenticator generates OTP	Enterprise login	No SMS dependency	Google Authenticator
34	Push-based MFA	Push notification	Enterprise	Best UX + security	Okta Verify

		approve login			
35	Hardware Key Login (FIDO2/U2F)	USB security keys	High security orgs	Best anti-phishing	YubiKey login
36	Passkeys Login (Modern Passwordless)	Device-based cryptographic keys	Latest apps	Future-ready auth	Google / Apple passkeys
37	Role-based Login (RBAC)	Login + roles decide access	Enterprise apps	Admin/user separation	ERP/CRM
38	Permission-based Login (ABAC)	Policy-based dynamic access	Big enterprises	Complex security	Cloud IAM systems
39	Login with Rate limiting	Prevent brute force attacks	Secure web apps	Security for public apps	Banking websites
40	Federated Login (Identity Federation)	Login via identity broker	Multi-app ecosystems	Multi-app login	Login via Okta, Azure AD
41	Guest Login	Temporary account without signup	Ecommerce/media apps	Quick onboarding	YouTube guest mode
42	Anonymous Login	App provides temporary token	Mobile apps	Try before signup	Firebase anonymous auth
43	Invite-only Login	Only invited users can create	Enterprise/teams	Controlled access	Slack invite-only workspace
44	Single Device Login	Only one device allowed	Secure apps	prevent shared accounts	Banking apps
45	Concurrent Session Login	Allow multiple sessions	Consumer apps	flexibility	Netflix (multiple devices)

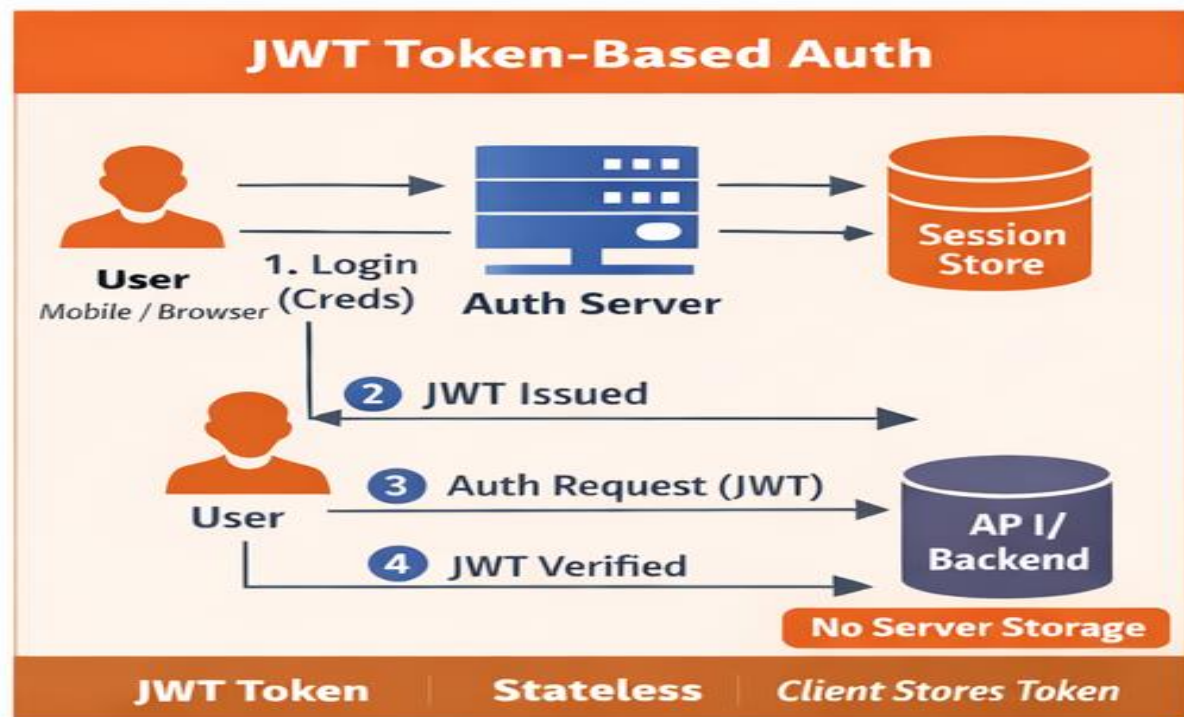
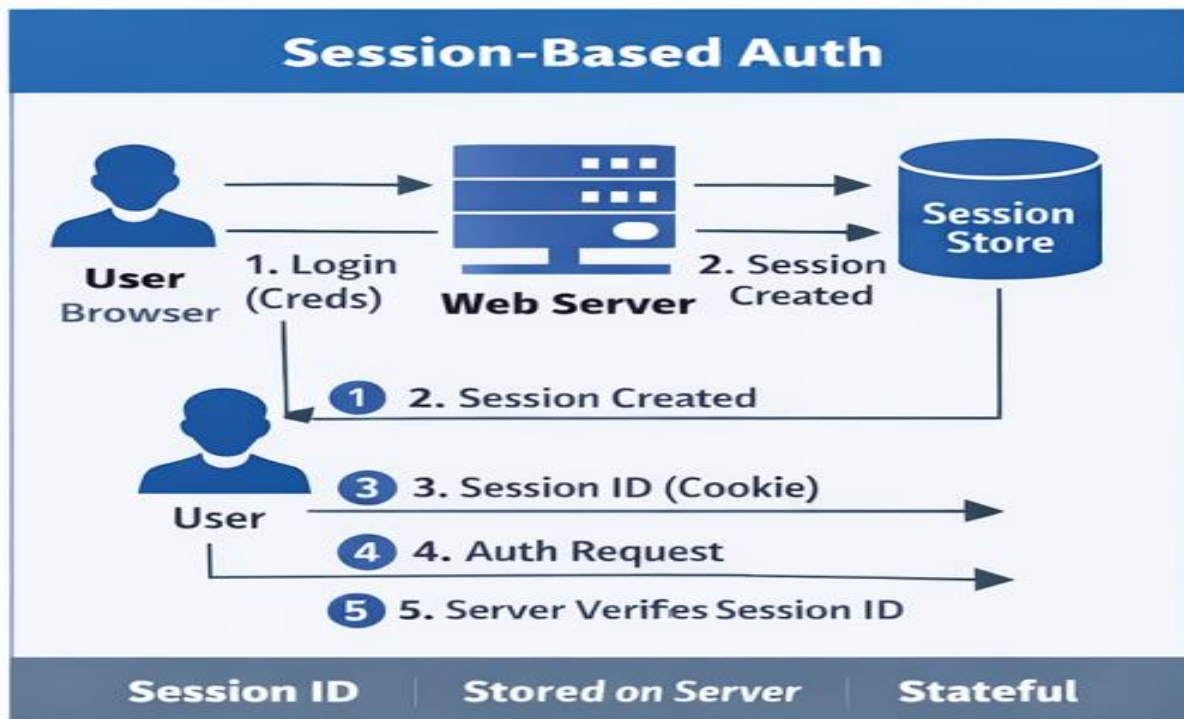
46	Admin Login + User Login separation	Separate auth flow	Enterprise software	Different security levels	Admin panel separate portal
47	Step-up Authentication	Extra verification for sensitive action	fintech apps	Balance security+UX	Login normal, pay requires OTP
48	Behavior-based Authentication	Analyze user behavior & risk	Banks	AI security	Fraud detection login
49	Risk-based Adaptive Login	Different auth based on risk score	Banking	dynamic security	Google suspicious login
50	Blockchain / Wallet Login	Login with wallet signature	Web3 apps	decentralized apps	MetaMask login

✔ Quick “When to Use What” (Short Guide)

Scenario	Best Login Method
Simple CRUD app / College project	Email+Password + Session
Mobile-first app	OTP login + JWT
Enterprise app	SSO (SAML / OIDC) + MFA
Banking / fintech	OTP + Biometric + Step-up auth
Developer APIs	API keys / HMAC
Web3 apps	Wallet signature login
Best modern experience	Passkeys

If you want, I can also provide:

- ✔ **Login implementation architecture diagrams** (Session vs JWT vs SSO)
- ✔ **Spring Boot + JWT login full project**
- ✔ **Django + OAuth + JWT full project**
- ✔ **Microservice login system (Auth service + Gateway + JWT + refresh token)**



Single Sign-On (SSO)



Single Identity | *Centralized Auth* | **SSO Token**